

Week 2 - Part A: Machine Learning Fundamentals

Introduction

Machine Learning (ML) is a field of Artificial Intelligence that allows computers to learn from data and make predictions without being explicitly programmed. Instead of following fixed rules, machine learning models find patterns in data and use those patterns to make decisions. In this report, I explain some basic machine learning concepts, including features and labels, data splitting, overfitting and underfitting, and the bias-variance trade-off.

1. Features vs. Labels

In machine learning, **features (X)** are the inputs given to the model, and **labels (y)** are the outputs that the model tries to predict.

For example, in the Titanic dataset:

Age = Feature

Gender = Feature

Passenger Class = Feature

Fare = Feature

Survived or Not Survived = Label

The model learns from the features and uses them to predict the label. Simply put, features are the information we provide, and labels are the answers we want the model to predict.

2. Train, Validation, and Test Split

A dataset is usually divided into three parts:

Training Set

The training set is used to teach the model. It learns patterns from this data.

Validation Set

The validation set helps check the model's performance during training and helps choose the best settings.

Test Set

The test set is used at the end to evaluate how well the model works on completely unseen data.

A common split is:

Training: 70%

Validation: 15%

Test: 15%

We keep the test data separate because it gives a fair measure of the model's performance. It is like a final exam that the student has never seen before.

3. Overfitting vs. Underfitting

Underfitting

Underfitting happens when a model is too simple and cannot learn the important patterns in the data. As a result, it performs poorly on both training and test data.

Overfitting

Overfitting happens when a model learns the training data too well, including noise and unnecessary details. It performs very well on training data but poorly on new data.

Ideal Model

A good model finds a balance between the two. It learns the important patterns without memorizing the training data.

4. Bias - Variance Trade-off

The bias-variance trade-off explains the balance between a simple model and a complex model.

High Bias usually leads to underfitting because the model is too simple.

High Variance usually leads to overfitting because the model learns too many details from the training data.

The goal is to find the right balance so the model can make accurate predictions on new data.

Conclusion

Machine learning uses features and labels to make predictions. Splitting data into training, validation, and test sets helps evaluate models fairly. Understanding overfitting and underfitting is important for building better models, and the bias-variance trade-off helps us find the right balance between simplicity and complexity.

What I Learned

I learned how machine learning uses features to predict labels.

I understood why data is split into training, validation, and test sets.

I learned the difference between overfitting and underfitting.

Issues I Faced & How I Solved Them

I was confused about validation and test data at first, so I looked at examples to understand their roles.

The bias-variance trade-off seemed complicated, but after reading about overfitting and underfitting, it made more sense.